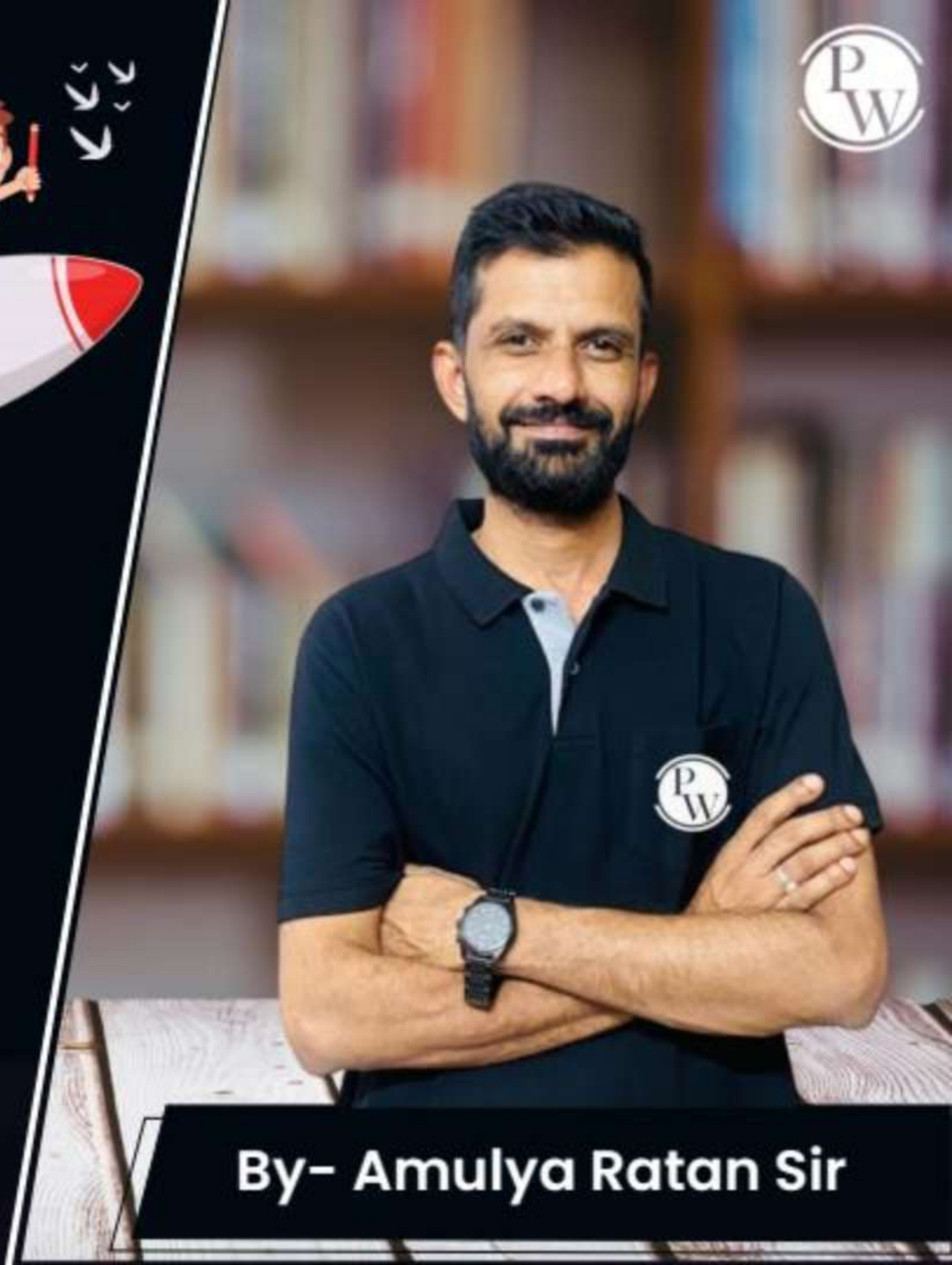
GATE

ALL BRANCHES

General Aptitude

QUANTITATIVE APTITUDE



Lecture No.- 02

By- Amulya Ratan Sir

Recap of Previous Lecture

Topic Calendars

① Date → Day?

② Calendar Representation

Topics to be Covered

Topic-1 Clocks

Topic-2 Averages

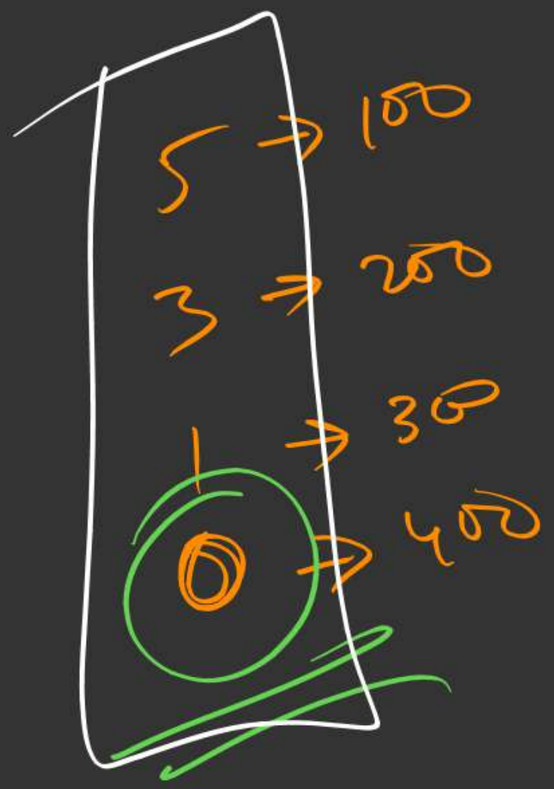
Topic-3 Percentages

Conditional Question

Saturday
Tuesday
Thursday

100 ⇒ 5 FRIDAY ✓
200 ⇒ 3 WEDNESDAY ✓
300 ⇒ 1 MONDAY ✓
400 ⇒ 0 SUNDAY ✓

Q



9) 400 (44
36
40
36
4

400 yrs

400

L.Y | N.Y.

100 | 300

−3 | +3

97 | 303

194 | 303

5 | 2

+ = 7

[MCQ]

#Q. Mahatma Gandhi was born on 2nd October 1869. What was the week Day?

A. Monday

B. Thursday

C. Tuesday

D. Saturday

Saturday

1800 ⇒ 3

68 ⇒ 1

L.Y 17 34

N.Y 51 51

6 + 2 = 8

= 6

1868 ⇒ 4 + 2

Jan - 3 July - 3

Feb - 0 Aug - 3

Mar - 3 Sep - 2

April - 2 Oct - 2

May - 3

June - 2

[MCQ]

#Q. The day of the March 16th of any year is the same day of the week as the corresponding date in which month of the same year?

A. July

B. November

C. April

D. September

16th March → 3

April → 2

May → 3

June → 2

July → 3

Aug – 3

Sep – 2

Oct – 3

16th Nov

[MCQ]

#Q. If January 1st, 1992 was a Wednesday. What day of the week was January 1st, 2003?

(A) Sunday

(B) Wednesday

(C) Thursday

(D) Friday

+0

1992 - 93 ⇒ 2

98 - 99 ⇒ 1

99 - 00 ⇒ 1

00 - 01 ⇒ 2

01 - 02 ⇒ 1

02 - 03 ⇒ 1

93 - 94 ⇒ 1

94 - 95 ⇒ 1

95 - 96 ⇒ 1

96 - 97 ⇒ 2

97 - 98 ⇒ 1

11 yrs

L.Y: 3, 6

N.Y: 8, 8

6 + 1 = 7

0

CLOCK

Relative speed

Relative Speed:

opposite $\rightarrow (+)$

Same $\rightarrow (\sim)$

Relative Speed:

1. Hands
2. Movement
3. Relative Speed

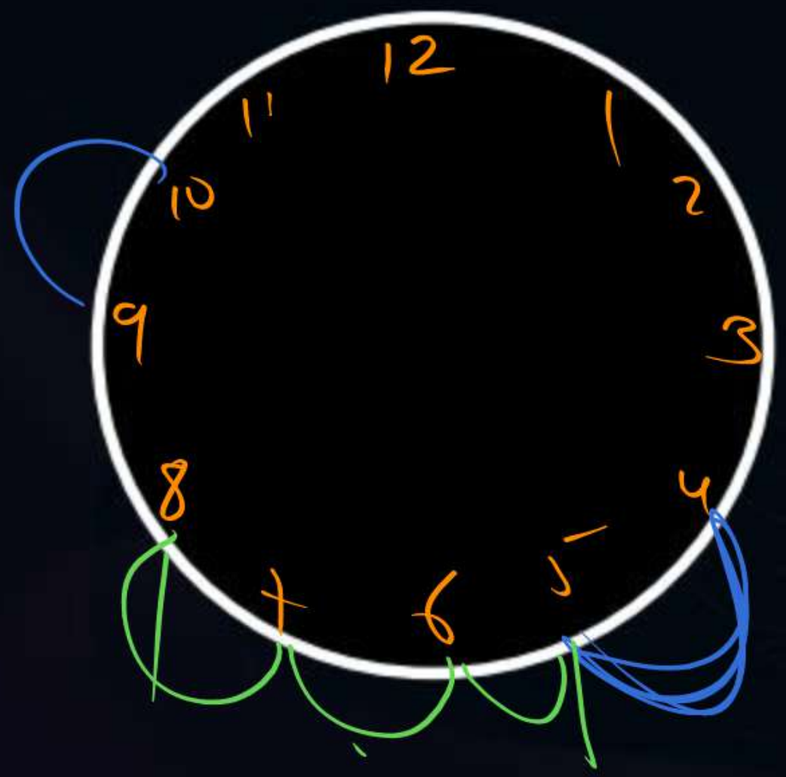
Same

(~)

Minimum

2

Dial of Clock



$$\frac{360^\circ}{12} = 30^\circ$$

(Minute Hand)

M.H

$\Rightarrow$

Dial of Clock

Relative Speed

5.5°/minute

M.H = $\frac{30}{5}$ = 6°/min

(Hour hand)

H.H = $\frac{30}{60}$

= $\frac{1}{2}$°/min

= 0.5°/min

To be Noted:

- ✓ Minute Hand Covers 30° in 5 minutes $30°/5 =$

  $6°/\text{min.}$

- ✓ Hour Hand Covers 30° in 60 minutes $30°/60$

  $= \frac{1}{2}° / \text{min}$

  or

  $0.5°/\text{min}$

5.5°/min

Different Patterns of Questions:

1. Time → Angle?
2. Angle → Time?
3. Gain OR Loss

R.S. = 5.5°/min

First Pattern:

TIME

$7 \times 30^\circ = 210^\circ$

7:00

4:00

↓

120°

5:00

150°

2:00

↓

60°

Random time given:

$\boxed{6:45} \rightarrow 67.5^\circ$

$6 \rightarrow 180^\circ$

$45 \times 5.5 \rightarrow 247.5^\circ$

67.5°

Questionnaire:

$345^\circ \rightarrow 5:30 \rightarrow 15^\circ$

$5 \Rightarrow 150^\circ$

$30 \times 5.5 \Rightarrow 165^\circ$

15°

Questionnaire:

352.5°

7.5°

3:15

$3 \rightarrow 90^\circ$

$15 \times 5.5 \rightarrow 82.5^\circ$

7.5°

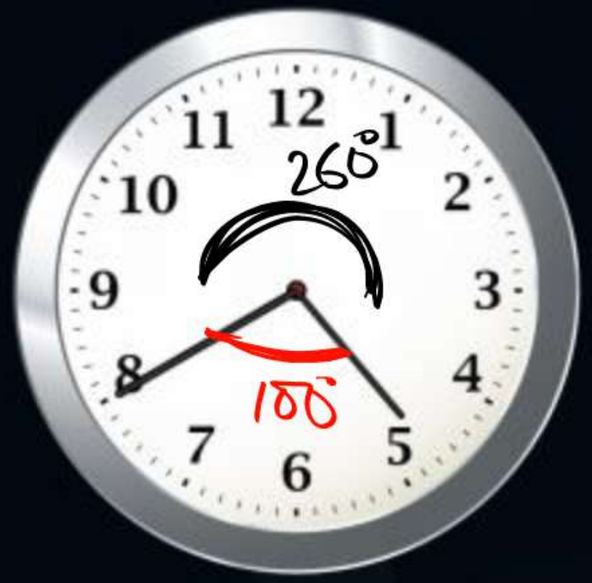
Questionnaire:

4:40 → 100°

4 → 120°

40 × 5.5 → 220°

100°



[MCQ]

#Q. At 9:45, the two hands of a clock make an angle of?

A. 60°

B. 45°

C. 33 ⅓°

D. 22 ½°

$9 \rightarrow 270^\circ$

$45 \times 5.5 \rightarrow 247.5$

22.5°

$360 - 22.5^\circ$

$= 337.5^\circ$

Two Answers:

Second Pattern:

Angle → Time?

0° OR Coincide:

12 hours → 11 times

24 hours → 22 times

180° OR Opposite:

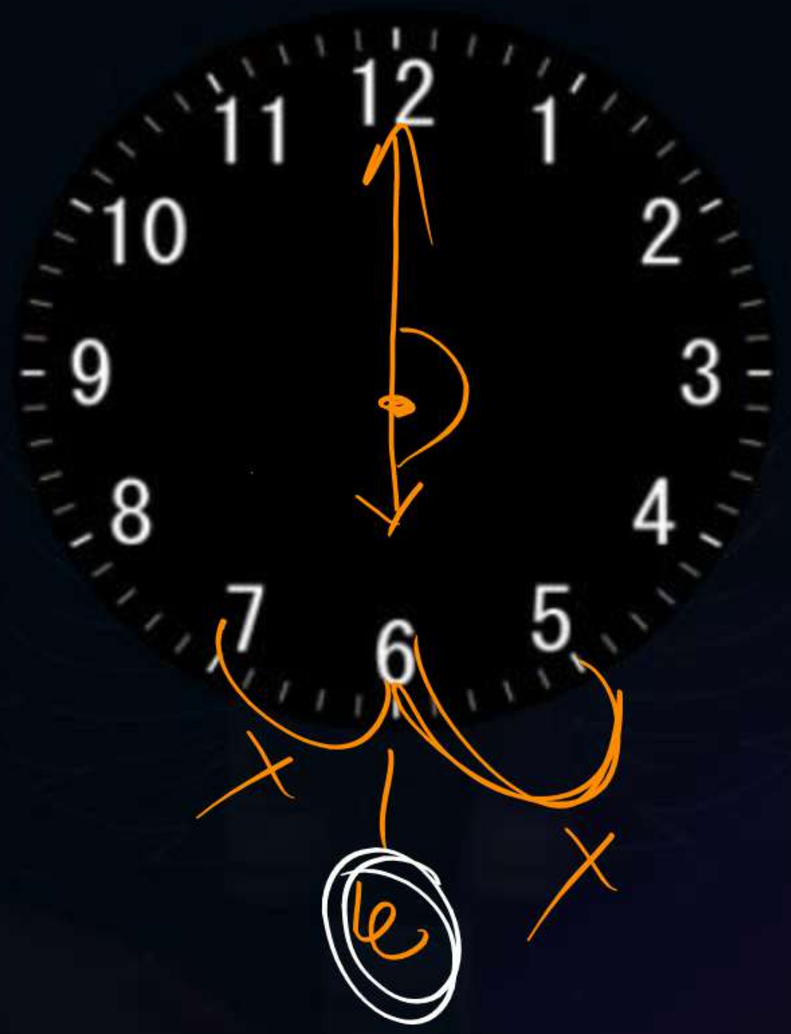
1 hours → 1 times

12 hours = 11 times

24 hours = 22 times

0° OR 180°

Except

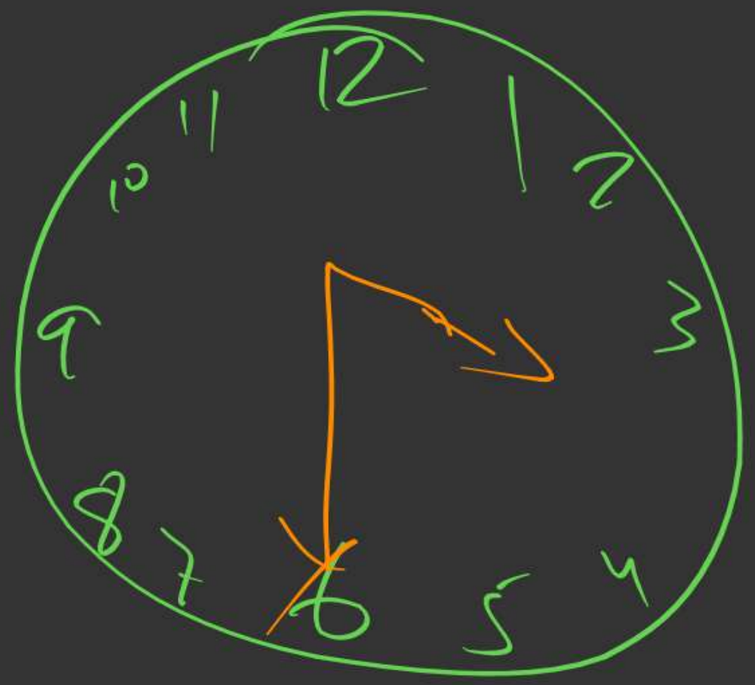


90° OR Right angle:

Any Angle except 0°&180°.

12 hours ⇒ 22 times

24 hours ⇒ 44 times



$3:30 \rightarrow 75^\circ$

$3 \rightarrow 90^\circ$

$30 \times 5.5 \rightarrow 165^\circ$

75°

Questionnaire:

#Q. In between 2 0' clock and 3 0' clock at what time the hands of clock form 90°?

$2 : 27\frac{3}{11}$

$\frac{150^\circ}{5.5} \times \frac{2}{2} = \frac{300}{11}$

$11 \overline{)300}\ 27$
22
80
77
3

$= 27\frac{3}{11}$ minutes

Questionnaire:

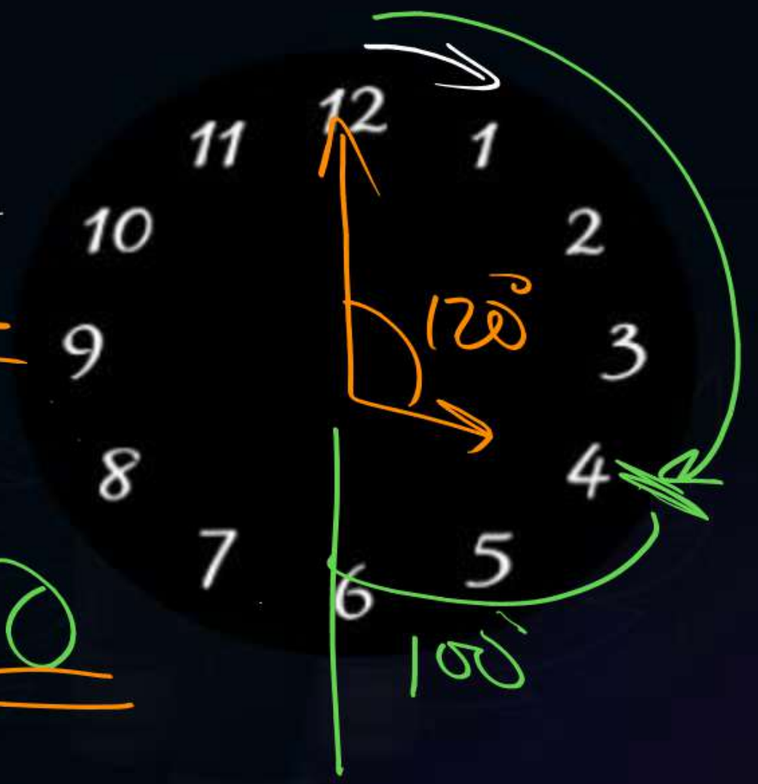
$4:03\frac{7}{11}$

$4:40$

#Q. In between 4 O' clock and 5 O' clock at what time the hands of clock form 100°?

$\frac{20^\circ}{5.5} = \frac{40}{11} = 3\frac{7}{11}$

$\frac{220}{5.5} = \frac{440}{11} = 40$



Questionnaire:

Two Answers

$6:43\frac{7}{11}$ $6:21\frac{9}{11}$

#Q. In between 6 O' clock and 7 O' clock at what time the hands of clock form 60°?

$$\frac{120}{5.5} = \frac{240}{11} = 21\frac{9}{11}$$

$$\frac{240}{5.5} = \frac{480}{11} = 43\frac{7}{11}$$

Questionnaire:

$1:23\frac{7}{11}$

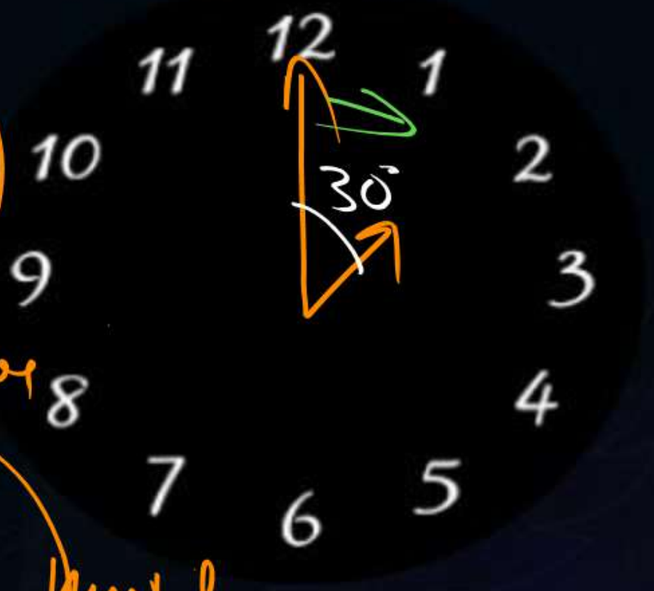
$1:52\frac{8}{11}$

#Q. In between 1 O' clock and 2 O' clock at what time the hands of clock form 100°?

$260°$

$\frac{130°}{5.5} = \frac{260}{11} = 23\frac{7}{11}$ minutes

$\frac{290°}{5.5} = \frac{580}{11} = 52\frac{8}{11}$ minutes



Questionnaire:

$1-2$ @ 100°

2PM

(−)

$$\frac{40}{5.5} = \frac{80}{11} = 7\frac{3}{11}$$ mm

$1:52\frac{8}{11}$

Third Pattern:

Gain

Lose

Questionnaire:

#Q. A Clock which gains 5 minutes in every one hour was set correct at 5am. What would be the time shown by that clock at 1pm the same day?

1:40

Questionnaire:

#Q. A clock which looses 10 minutes in every one hour was set correct at 4am, what would be the time shown by that clock at 4pm the same day?

(−) 120 min

2PM

2 mins Summary

Topics

Clocks, ~~Averages, Percentage~~

THANK - YOU